

BDIC1032J Advanced Mathematics (Module 4; Engineering)

SECTION A — GAP-FILLING QUESTIONS

This section is worth a total of **60** marks, with each question worth **4** marks.

1. Determine the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{\sqrt[3]{xy^2+1}-1} = \underline{\hspace{2cm}}$
2. Given the function $z = xy + f(x^2 - y^2)$, where $f : \mathbb{R} \mapsto \mathbb{R}$ is differentiable, compute $y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} = \underline{\hspace{2cm}}$
3. Suppose that the relation $e^z - xyz = 0$ defines an implicit function $z = z(x, y)$. Then $dz = \underline{\hspace{2cm}}$
4. Find equations of the tangent lines to the plane curve $x = 2t^3 + 1$, $y = 3t^2 + 1$ that pass through the point $(3, 4)$. *Hint:* There are two such tangent lines.
Answer for the tangent lines: $\underline{\hspace{2cm}}$; $\underline{\hspace{2cm}}$
5. The intersection of the two surfaces $x^2 + 2y^2 - z^2 + 3x = 1$ and $2x^2 + 4y^2 - 2z^2 - 5y = 0$ lies in a plane. Find an equation of the plane: $\underline{\hspace{2cm}}$
6. Find the directions in which the function $f(x, y) = \sin(xy) + y^2$
 - (a) increases most rapidly at the point $P(1, 0)$. Answer: $\underline{\hspace{2cm}}$
 - (b) has zero directional derivative at the point $P(1, 0)$. Answer: $\underline{\hspace{2cm}}$
7. Find the points on the hyperboloid $x^2 + 4y^2 - z^2 = 4$ where the tangent plane is parallel to the plane $2x - 2y + z = 5$. Answer for the points: $\underline{\hspace{2cm}}$

BDIC1032J Advanced Mathematics (Module 4; Engineering)

8. The plane $x - y + 2z = 2$ intersects the paraboloid $z = x^2 + y^2$ in an ellipse. Find the points on the ellipse that are nearest to and farthest from the origin.

Answer for the nearest point: _____ Answer for the farthest point: _____

9. A rectangular box without a lid is to be made from 12 m² of cardboard. Find the maximum volume of such a box. Answer for the maximum volume is _____.

10. Calculate the iterated integrals $\int_0^1 \left(\int_{\sqrt{y}}^1 \sqrt{x^3 + 3} dx \right) dy =$ _____

11. Suppose that $f(x, y) + \iint_D f(x, y) dx dy = \sqrt{a^2 - x^2 - y^2}$, and $D = \{(x, y) \mid x^2 + y^2 \leq a^2\}$. Then $f(x, y) =$ _____ $\iint_D f(x, y) dx dy =$ _____

12. The three-dimensional region Ω is such that the triple integral $\iiint_{\Omega} (4 - 3x^2 - 2y^2 - z^2) dx dy dz$ achieves the maximum possible value. Then

Ω is _____ and the maximum integral equals _____

13. Let C be the plane curve $|x| + |y| = 1$. Use the Green's theorem to evaluate the line integral $\oint_C (3y - \ln(\cos^2 x + 1)) dx + (5x - \sqrt{y^4 + 1}) dy =$ _____

14. Use symmetry to compute the double integral

$$\iint_{x^2 + y^2 \leq 1} (x^2 + e^{-x^2} \sin y + y^2 \tan^3 x + 1) dx dy =$$

15. The gravitational force field $\mathbf{F}(x, y, z) = -\frac{mMG}{|\mathbf{x}|^3} \mathbf{x}$, $\mathbf{x} = \langle x, y, z \rangle \in \mathbb{R}^3$, $\mathbf{x} \neq \mathbf{0}$, is a conservative vector field satisfying $\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}$. Find a scalar function $f(x, y, z)$ such that $\mathbf{F} = \nabla f$.

Answer for the potential function: $f(x, y, z) =$ _____

BDIC1032J Advanced Mathematics (Module 4; Engineering)

SECTION B — EXTENDED ANSWER QUESTIONS

This section is worth a total of **40** marks, with each question worth 10 marks.

- 16. (10 marks)** First use polar coordinates to evaluate the double integral $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)/2} dx dy$, and then use this result to determine the single integral $\int_0^\infty e^{-x^2/2} dx$.
- 17. (10 marks)** Find the volume of the solid Ω that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = a^2$.
- 18. (10 marks)** Evaluate the line integral $\oint_C y^2 dx + 3xy dy$, where C is the positively oriented boundary curve of the semi-annular region D in the upper half plane between the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$.
- 19. (10 marks)** Use the divergence theorem to evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = \langle x^2 + e^{yz^2}, xy, xy \sin(xy) \rangle$ and S is the surface of the solid region E bounded by the parabolic cylinder $z = 4 - y^2$ and the planes $z = 0$, $x = 0$ and $x + z = 6$.

BDIC1032J Advanced Mathematics (Module 4; Engineering)

20. (Bonus problem: 20 marks) Consider the electric field $\mathbf{E}(\mathbf{x}) = \frac{\varepsilon Q}{|\mathbf{x}|^3} \mathbf{x}$ where the electric charge Q is located at the origin and $\mathbf{x} = \langle x, y, z \rangle \in \mathbb{R}^3$, $\mathbf{x} \neq \mathbf{0}$, is the position vector.

- (a) Let S be a sphere centered at the origin and with radius $a > 0$. Verify directly that the electric flux of $\mathbf{E}$ through S satisfies $\iint_S \mathbf{E} \cdot d\mathbf{S} = 4\pi\varepsilon Q$.
- (b) Show that the electric flux of $\mathbf{E}$ through *any* closed surface S that encloses the origin also satisfies $\iint_S \mathbf{E} \cdot d\mathbf{S} = 4\pi\varepsilon Q$.